

SENTIMENT ENGAGEMENT ANALYZER

Project Summary — Derek Zhou — February 18, 2026 —
<https://github.com/dezh820-awesome/Stock-Prediction-Project>

WHAT IS IT?

A modular Python framework that predicts Tesla daily stock returns by combining FinBERT sentiment features from Elon Musk's tweets with OHLCV technical indicators. Two nested models are compared:

- Baseline — OHLCV only (SMA, RSI, MACD, Bollinger Bands, ATR, OBV, volume)
- Sentiment-enhanced — baseline plus rolling sentiment scores, engagement-weighted ratios, Shannon entropy diversity, and lag features

Tesla was chosen for its strong coupling between executive communication and market response, providing a controlled setting to examine concentrated sentiment influence.

KEY RESULTS

Metric	Baseline	Enhanced	Δ
Test R^2	0.722	0.727	+0.005
RMSE	0.042	0.040	-0.002
MAE	0.030	0.029	-0.001
CV R^2	0.718	0.724	+0.006

Nested F-test: $F = 12.47$, $p = 0.032$. Bootstrap 95% CI for ΔR^2 : $[0.002, 0.008]$ — excludes zero. Top predictors: `positive_ratio` ($p < 0.001$) and `sentiment_diversity` ($p < 0.05$).

The high absolute R^2 reflects contemporaneous technical features and volatility clustering. The primary finding is the *incremental* contribution of sentiment.

HOW IT WORKS

1. Tweets preprocessed and classified by FinBERT (confidence ≥ 0.60)
2. Daily features: counts, ratios, rolling means (3/7/14-day), engagement weighting, Shannon entropy, lag features
3. Technical indicators from DataBento OHLCV (Jan 2010 – Oct 2025)
4. Temporal 80/20 split: training 2010–2021, testing 2022–2025
5. Ridge regression with `TimeSeriesSplit` CV; scaler fit inside each fold
6. Full LINE diagnostics + nested F-test confirm statistical validity
7. Conformal prediction intervals calibrated from training residuals

OPEN QUESTIONS

- Does the +0.5% R^2 gain translate into economically meaningful trading performance? A Sharpe ratio analysis with transaction costs has not been done.
- How much of the signal is attention (tweet volume) vs. sentiment content (positive/negative tone)?
- Do the results generalise to other equities, or are they specific to Tesla's media-driven valuation dynamics?
- Would nonlinear models (Random Forest, XGBoost, LSTM) extract additional structure that linear regression misses?
- How quickly does the sentiment signal decay? Is there a detectable structural break in coefficient stability over time?
- Are gains driven by a few high-impact events (earnings, product launches) rather than day-to-day sentiment?
- Would expanding to multiple sources (news, Reddit, other executives) reduce representational bias and improve generalisability?

LIMITATIONS

- Single equity and single sentiment source
- Linear models only — nonlinear dynamics not captured
- FinBERT may misclassify sarcasm, memes, and humour
- Statistical significance does not imply economic significance
- Winsorisation applied to full sample rather than training data only

ETHICAL NOTE

Models trained on single-individual sentiment data may reinforce power asymmetries in financial markets. Sentiment may reflect attention rather than fundamental value, and predictive utility is likely transient. Responsible deployment requires transparency, regular auditing, and avoidance of overconfident automated trading.

DATA SOURCES

- DataBento `ohlcv-1d` — Tesla OHLCV, Jan 2010 – Oct 2025
- Kaggle — Elon Musk Tweets 2010–2025 (~50,000 tweets)
- FinBERT `yyanghkust/finbert-tone` — sentiment classification